final_project_code

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1. Reading in the data

```
read_data <- read_excel("./stat571B_reading_data.xlsx")

read_data['total_time'] = read_data['Time (minutes)']*60 + read_data['Time (seconds)']

read_data['format'] = c("Laptop", "Laptop", "Print",
                        "Laptop", "Print", "Print",
                        "Print", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Laptop", "Print", "Print",
                        "Laptop", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Print", "Laptop", "Print",
                        "Laptop", "Print", "Laptop",
                        "Laptop", "Laptop", "Print",
                        "Print", "Laptop", "Print",
                        "Print", "Print", "Laptop")

read_data["music"] = c("no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no")

read_data
```

```
## # A tibble: 36 x 8
##    Participant  Text `Comprehension score (%)` `Time (minutes)` `Time (seconds)`
##    <chr>       <dbl>                     <dbl>            <dbl>            <dbl>
##  1 P1              1                       100                2               17
##  2 P1              2                        75                2               45
##  3 P1              3                        75                1               58
##  4 P2              1                        75                3               14
##  5 P2              2                        50                2               36
```

```
##  6 P2              3                        75                2                51
##  7 P3              1                        75                2                 5
##  8 P3              2                       100                2                28
##  9 P3              3                        75                1                49
## 10 P4              1                        75                2                22
## # i 26 more rows
## # i 3 more variables: total_time <dbl>, format <chr>, music <chr>
```

2. Analysis

We used ANOVA with participant as a blocking factor to assess the significance of the experimental factors, as required. Additionally, linear mixed-effects models were fitted with participant as a random effect to account for within-subject correlation.

We fit factorial models with format, music, and their interaction as primary fixed effects. Because participants each read multiple passages, participant was included to account for within-subject variability. Since only three text passages were used, Text was included as a fixed effect rather than a random effect to adjust for possible passage-specific differences.

Model 1: Time

```r
# ANOVA for Reading Time
model_time <- lm(total_time ~ format * music + Text + Participant, data = read_data)
Anova(model_time, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: total_time
##              Sum Sq Df F value    Pr(>F)
## (Intercept)   43035  1 85.6730 1.801e-08 ***
## format          286  1  0.5697 0.4596296
## music          1688  1  3.3601 0.0825111 .
## Text            956  2  0.9518 0.4037164
## Participant   35792 11  6.4776 0.0002125 ***
## format:music     27  1  0.0535 0.8195047
## Residuals      9544 19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Mixed Effects Model (Supplementary Analysis)
lmer_time <- lmer(total_time ~ format * music + Text + (1 | Participant), data = read_data)
anova(lmer_time)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF F value  Pr(>F)
## format        354.6   354.6     1 19.829  0.7078 0.41019
## music        4031.4  4031.4     1 19.829  8.0484 0.01024 *
## Text          956.2   478.1     2 19.107  0.9545 0.40263
## format:music   18.0    18.0     1 19.829  0.0359 0.85155
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For reading time, neither format nor the interaction between format and music was statistically significant. Music showed a marginal effect in the ANOVA model and was statistically significant in the mixed-effects model, suggesting a potential influence of music on reading time after accounting for within-subject variability.

The effect of Text was not statistically significant, indicating no strong evidence of differences in reading time across the three passages.

Model 2: Comprehension

```r
# ANOVA for Comprehension
model_comp <- lm(`Comprehension score (%)` ~ format * music + Text + Participant, data = read_data)
Anova(model_comp, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: Comprehension score (%)
##               Sum Sq Df F value    Pr(>F)
## (Intercept)  17259.0  1 62.9608 1.892e-07 ***
## format         850.7  1  3.1033   0.09422 .
## music          850.7  1  3.1033   0.09422 .
## Text           138.9  2  0.2533   0.77879
## Participant   2986.1 11  0.9903   0.48795
## format:music  1250.0  1  4.5600   0.04596 *
## Residuals     5208.3 19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

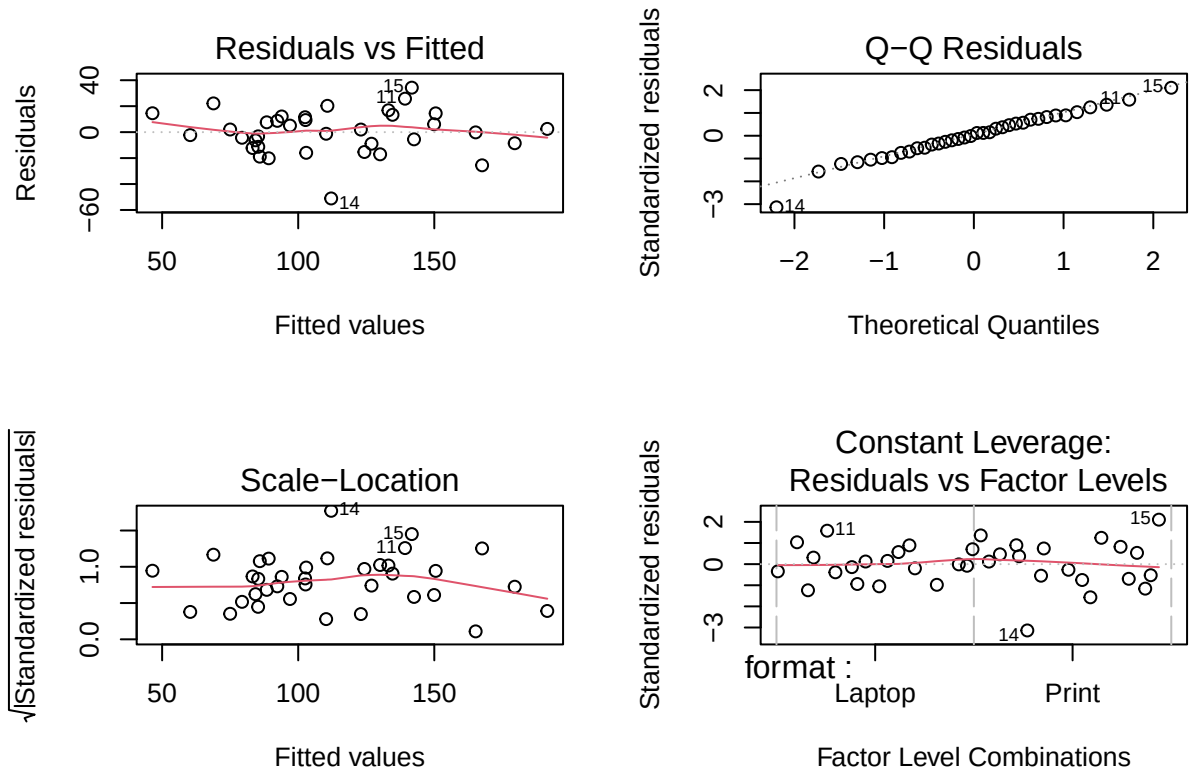
```r
## Mixed Effects Model (Supplementary Analysis)
lmer_comp <- lmer(`Comprehension score (%)` ~ format * music + Text + (1 | Participant), data = read_da
anova(lmer_comp)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF  DenDF F value  Pr(>F)
## format         69.06   69.06     1 23.714  0.2537 0.61910
## music           0.00    0.00     1 23.714  0.0000 0.99737
## Text          138.89   69.44     2 19.277  0.2551 0.77738
## format:music 1112.33 1112.33     1 23.714  4.0867 0.05465 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

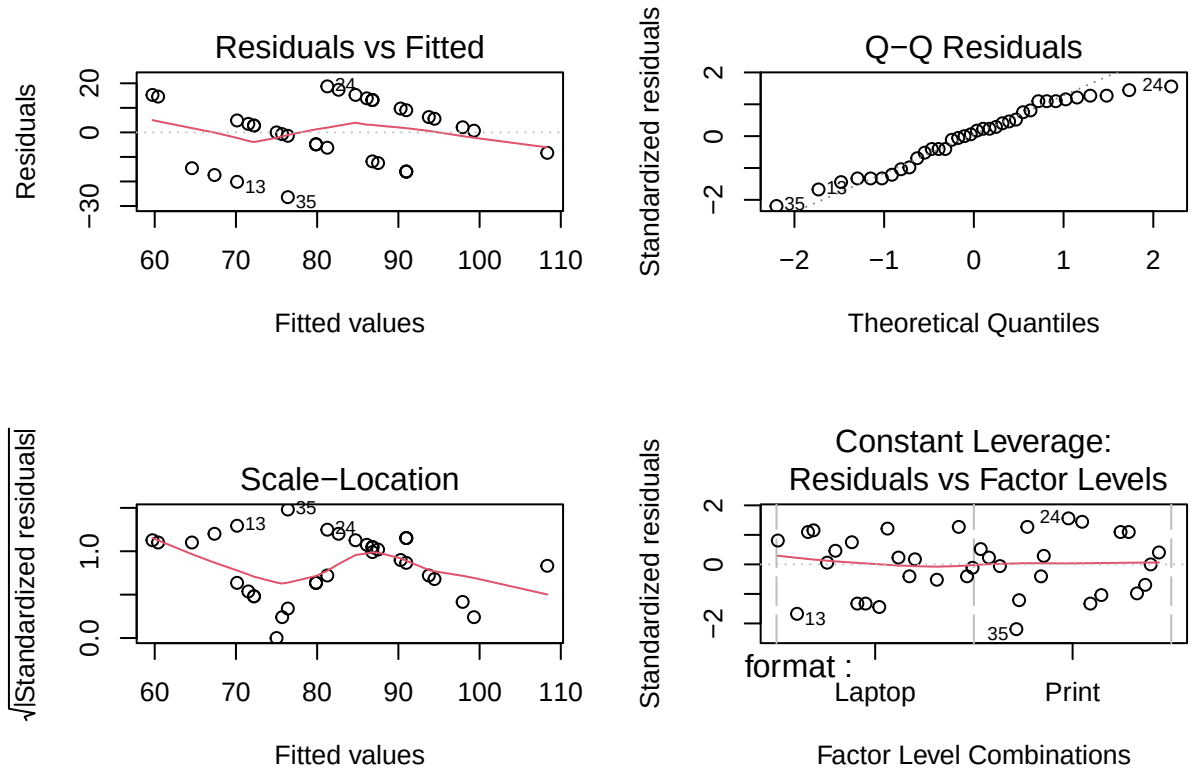
For comprehension, the interaction between format and music was statistically significant in the ANOVA model and marginally significant in the mixed-effects model. This indicates that the effect of format on comprehension depends on the presence of music, and therefore the main effects of format and music should not be interpreted independently. The main effects of format and music were not statistically significant on their own. The effect of Text was also not statistically significant, suggesting no strong evidence of differences in comprehension across passages.

Diagnostics for ANOVA

```r
par(mfrow = c(2,2))
plot(model_time)
```




```
par(mfrow = c(2,2))
plot(model_comp)
```



Diagnostic plots for both models indicate no major violations of model assumptions.